

A Pre-feasibility Study for the Establishment of a Dairy Products Factory



2017



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1. Executive Summary

The project aims to establish a factory for the manufacturing of milk and dairy products in AlTafilah governorate, and this project is a unique project on the scale of the governorate especially as it suffers from a shortage of those products.

The project, if established, will raise the level of the industrial sector in the governorate and encourage investment in it. Based on the proposed market share and the prices that we will be presenting later in detail, which were presented by the proposed project, the expected revenues amounted to 3,246,336 JD during the first year of the project and gradually increase to 5,140,032 JD during the tenth year. The period of capital recovery is 5 years with an internal rate of return equal to 26.13%, in addition to providing 35 jobs.

The following table shows the financial indicators of the project:

Table1 economical indicators

Indicator	Value
Total Investment Cost (Jordan Dinars)	2,512,005
Internal Rate of Return (IRR)	%26.13
Pay-Back Period (Years)	5
Net Present Value (Jordan Dinars)	2,447,658
jobs opportunities that provided by the project	35
Expectations annual revenue (for 10 years)	41,931,840
Total operating costs (for 10 years)	30,423,209
Gross profit (for 10 years)	5,014,754

Key Highlight of Jordan

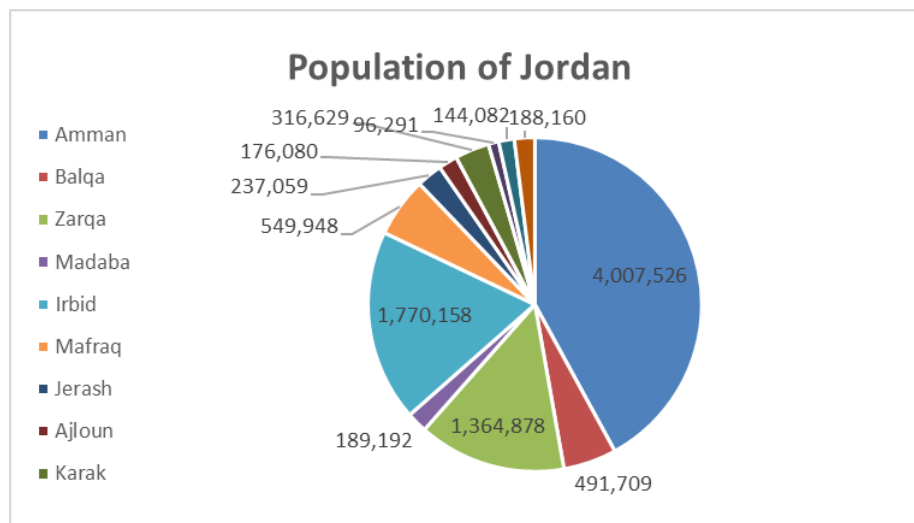
Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.



of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafrq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table2 Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

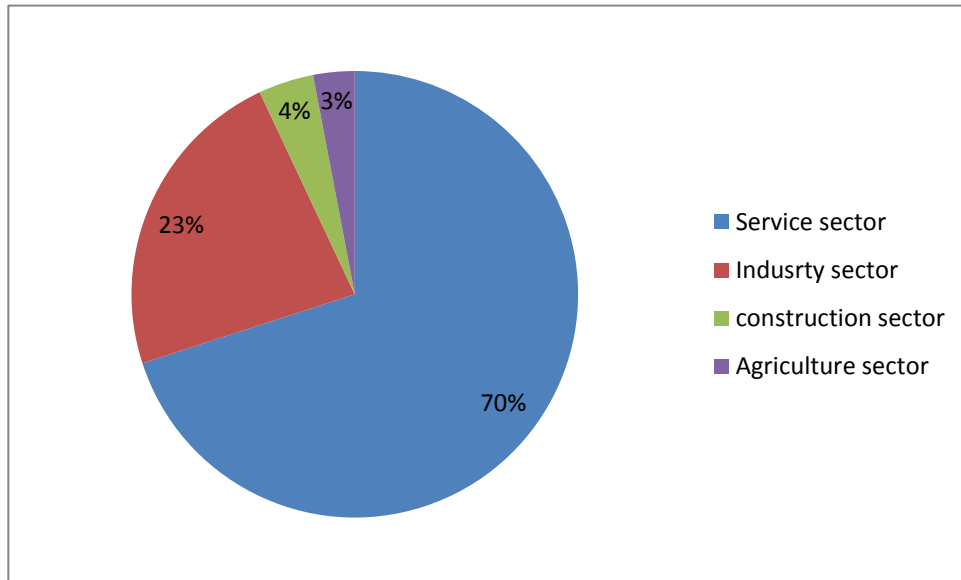
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan

government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance. The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

Furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from

customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

Investment Climate in Al-Tafielh:

Under the provisions of fifth article of the investment law No. 30 of 2014 and Income tax regulations in the less developed areas, the reduction of the income tax system on the investment projects are clarified in the table below:

Table3 Income Tax and Exemptions

Within the Development Zone or Free Zone	Less Developed Areas
Income Tax is 5%	Income Tax is 20%
	As per Investment Law categorization, Irbid is classified in Class (B) where investors enjoys 80% exemption on Income tax for 20 years, where net income tax after reduction

2. Market analysis

Project Description

The project aims to establish a plant in AlTafilah governorate to produce milk and dairy products. The idea of establishing the project is considered a good idea especially, when such products are widely sold in markets and supermarkets because of their nutritional benefits and are recommended for adults and young people and can be easily marketed.

The production capacity of the plant is estimated to be about 800 tons / year of fresh pasteurized milk used for manufacturing various dairy products such as: milk, yogurt, white cheese, Labaneh and Jameed.

Raw Materials Requirements

The required raw materials are summarized in the table below:

Table4 raw material required

Raw materials	Measuring unit	Cost (JD)
Fresh raw milk	L	0.27
Sugar	Kg	0.36
Salt	Kg	0.17
Rennet	G	0.13
Calcium chloride	Kg	0.51
Stabilizers	Kg	10.68
Allowed artificial colors	Kg	20.5
Plastic containers	1000 package	17.10
Aluminum covers	1000 cover	11.10
Cartoon paper	1000 one liter package	111
Plastic bags	bags 1000	85.40
Plastic boxes	box	1.30

Project Objectives

The objectives of the project are summarized as follows:

- Encouraging investment in the industrial sector in the governorate.
- Supplying the local market with essential dairy products.
- Providing job opportunities and improving the living and social standards of the workers in the project.
- Achieving financial return for the owners.

Proposed Products

The proposed project will establish a factory for the manufacturing of milk and dairy products and distribute them to the markets within the governorate to cover the demand of dairy products, especially as it suffers from a shortage of those products because of their geographical distance from food distribution companies. In addition, it will serve the markets of other southern governorates.

The plant is expected to produce several types of dairy products, and the following table illustrates those products:

Table5 the proposed products

products
Pasteurized milk
Jameed
Yogurt
Labneh
White cheese

Target segments

The project mainly targets local markets within the governorate to provide them with dairy products, and also targets the families and residents of the governorate by providing them with job opportunities, as well as supplying dairy products to southern governorates.

Market Size Analysis

Based on the recorded economic statistics, the total production of the dairy industry in the Kingdom reached 537.52 tons during the seven months from January to July 2016. The amount of imported milk products for the year 2015 reached (6,487 tons), while the quantity of exports of the same product reached to (979 tons), this means that the rate of imports is higher than the rate of exports, which reinforces the idea of establishing factories and support the local industrial sector. The following table shows the Kingdom's import and export rate for dairy products from 2011-2015.

Table 6 rate of Imports and exports of dairy 2011-2015

Years	Value of imports (MJD)	The amount of imports (Ton)	National exports (MJD)	The amount of exports (Ton)	Value of exports (MJD)	Value of trade deficit (JD)
2015	10,237,442	6,487	1,946,699	979	2,464,081	-7,773,361
2014	12,855,964	7,520	1,281,378	531	1,314,364	-11,541,600
2013	10,078,444	6,508	2,173,927	1,723	2,397,822	-7,680,622
2012	4,613,250	3,537	3,140,380	2,970	3,483,745	-1,129,505
2011	9,912,335	9,941	1,538,972	1,360	1,875,362	-8,036,973
Average	9,539,487	6,799	2,016,271	1,513	2,307,075	-7,232,412

Department of Statistics http://www.dos.gov.jo/dos_home_a/main/linked-html/external_trade.htm

Competitors Analysis

Based on market studies and analysis by the team, there are direct and indirect competitors: Direct competitors, Hamouda Food Industries Company, where it contains an official agent within the governorate and distributes its products to commercial markets.

Indirect competitors such as Baladna Food Industries and Al-Mazra'a for food processing. They distribute their products from the capital to the governorate within a specific distribution mechanism, in addition to other food companies such as Al-Youm and Taiba Food Industries.

Price Analysis and Pricing Policy

Pricing policy is an important component of the marketing, which determines the project's ability to attract targeted segments. The prices of dairy products are fairly stable, and the following table shows the prices of the factory's products compared to their prices in the local markets:

Table7 Factory product prices compared with the markets (JD)

Products	Units	Market price	Factory price
Pasteurized milk	L	1.25-1	1
Al-Jameed	Kg	26-24	25
Yoghurt	Kg	0.75-0.60	0.70
Labneh	Kg	5-3.50	3
White cheese	Kg	6-4	5

Market Share

According to the general statistics, the population of AlTafilah governorate is (96,291), and has a population density of 43.6 persons / km². The following table shows the average annual consumption per capita in AlTafilah governorate compared with the Kingdom for dairy products for 2013:

Table 8 average annual consumption per capita of dairy

Products	Consumption per person / year	
	AlTafilah	Kingdom
Pasteurized milk	0.62	1.33
Fresh milk	0.93	1.93
Yoghurt (Kg)	1.24	12.44
Normal buttermilk	1.41	3.46
Labneh (Kg)	2.08	3.41
White cheese	1.19	1.79

Department of statistics– <file:///C:/Users/n/Downloads/2013%207-1.pdf>

Based on the above, it was found that the governorate consumption rate of the products to be manufactured is (719) tons / year, and for the Kingdom amounted to 235,560 tons / year.

Projected Revenues

Projected Annual Revenues for the Project

Based on the studies and assumptions of the Jordanian market size in the amount of consumption of milk and the analysis of the pricing policy of factory products, it was found that the revenue of the factory in the first year of its establishment is (3,246,336) JD to reach the tenth year of the study to (5,140,032) JD. The following table shows the revenues of the plant within ten years from the beginning of the project:

Table 9 projected revenues

Price assumption	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
The production rate of full production	60%	65%	70%	75%	75%	80%	80%	85%	90%	95%
The selling price of pasteurized milk	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
The selling price of fresh Jameed	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00
The selling price of yoghurt (Kg\JD)	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70
The selling proce of Labneh (Kg\JD)	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00
The selling price of white cheese	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00
Revenues of pasteurized milk	92736	100464	108192	115920	115920	123648	123648	131376	139104	146832
Revenues of Jameed	2318400	251160	270480	289800	289800	309120	309120	328440	347760	367080
Revenues of yoghurt	67200	72800	78400	84000	84000	89600	89600	95200	100800	106400
Revenues of Labenh	288000	312000	336000	360000	360000	384000	384000	408000	432000	456000
Revenues of white cheese	480000	520000	560000	600000	600000	640000	640000	680000	720000	760000
Total revenues	3246336	351686	378739	405792	405792	432844	432844	459897	486950	514003

The following table shows the analysis of the production capacity of the plant, where the total capacity was imposed about (800) tons / year, and the proportion of each product is 20% of the total production capacity.

Table10 Analysis of the productive capacity of the plant

Production capacity	
Annual production capacity (Kg)	800000
Production rate of pasteurized milk	20%
Production rate of Jameed	20%
Production rate of yoghurt	20%
Production rate of Labaneh	20%
Production rate of white cheese	20%
The convert coefficient (kg to liter)	0.966

3. Technical Analysis

The study aims to develop the technical specifications of the project, based on the results of the market study, in addition to evaluating and reviewing investment and operational costs. The technical study of the project includes an explanation and a general description of the aspects of the project along with its components, in addition to the constructional and operational requirements of the project.

Required Human Resources

This project requires a factory manager who oversees all operational matters in the factory, as well as a financial manager to develop financial management plans and regulations, an accountant who manages the financial affairs of the factory, as well as production line technicians and cleaners to keep all facilities clean.

Table 11 the required labor for the project

Job title	No.
General Manager	1
Factory manager	1

Job title	No.
Financial Manager	1
Quality manager	1
Human resources manager	1
Quality auditors	2
Accountant	3
Secretaries	1
Human resources staff	2
Guard	2
Total of indirect staff	15
Production line technicians	10
Logistics Manager	1
Logistics staff	2
Maintenance technicians	2
Driver	5
Total of direct staff	21
Total	35

Job Description:

- General Manager:** perform all the tasks and responsibilities of the general manager, and active participation in the identification and formulation of objectives, as well as planning and organizing workflow facility to ensure the achievement of specific objectives.
- Factory manager:** Is working solo and within his authority, while his hiring by the person who owns the project. Also is propose policies and action plans and to represent the factory and workflow planning and management at the plant, following up the implementation of plans and programmes in accordance with determined policies and objectives of the plant optimally to achieve the interests of employers, and overseeing the management of administrative and financial affairs in accordance with prescribed instructions in this regard, managing subordinates, develop their skills, develop procedures and ensure occupational safety and health requirements.

- **Financial manager:** Define the organizational objectives, plan, organize and follows up on the implementation of the financial management plans according to the internal regulations of the factory and propose the appropriate annual budget for the plant within.
- **Quality manager:** Planning and organizing the work in accordance with ISO 9001 - 2015 and effective participating in the identification and formulation of objectives and performance of all tasks.
- **Human resources manager:** Develop procedures, policies and plans for the recruitment of employees, and follow-up on all with workers through services and administrative affairs and the provision and improvement of working methods.
- **Quality auditors:** Check the effectiveness of administrative performance and work procedures in accordance with the legislations in force.
- **Accountant:** follows-up financial matters and financial audit, and submit annual reports of the financial situation.
- **Secretary:** Receive the factory's correspondences and transactions and submit them to the factory manager, and also support the administrative staff.
- **Production line officer:** Following-up and monitoring of the production process according to pre-set plans.
- **Logistics Manager:** Planning, managing and supervising the plans, policies and procedures of the logistics services and providing the necessary consultations to support and implement the agreed strategic objectives, in addition to developing logistics services, which includes maintenance and services.
- **Maintenance technician:** implement actions and activities of electrical and mechanical maintenance, and follow up on the implementation of plans and programs planned in order to ensure the continuity of the production process and quality at full capacity with the lowest possible cost, and ensure the availability of spare parts, and the preparation of technical reports.
- **Driver:** Responsible for the transportation of products and distributing them to the markets.
- **Security:** Implement the approved security instructions to maintain the security system in the factory.

Site Analysis

It is proposed that the plant should be located on a land outside the organization on an area of 10,000 square meters and away from any source of microbial contamination, toxins or unpleasant odors, such as the livestock market, oil refineries, open sewage or others, in addition to the availability of water in appropriate quantities and quality. If possible, with a possible disposal of wastes. The following points should be considered:

- Taking into account future expansion possibilities.
- Isolate management sections and laboratories from production sections, also should be in separate buildings and production lines from each other with wide dividers
- Placing the tanks of raw milk in a high place to avoid frequent pumping.
- Building warehouses and stores at the level of the production lines to facilitate the entry of materials and finished products.
- Working conditions should be good within the factory, such as lighting, heating, ventilation, cleaning and other.

Departments of the Factory

Dairy plant should consist of the following:

- **Department of receiving raw milk:** In this department, the acidity of the milk is determined by which if the milk is accepted or rejected, milk is then weighted, refrigerated and stored in large tanks until processing.
- **Production Department:** It is the largest department in the factory, consisting of several production lines.
- **Storage Department** consists of two parts:
 - The First, consists of several large refrigerated rooms to store ready-made dairy products after they are manufactured until they are marketed.
 - The second, consists of several non-refrigerated warehouses for storing auxiliary materials in production, such as packaging materials from sterile milk bottles, plastic or metal packaging, and other production requirements.

- **Laboratories:** equipped with all the necessary equipment, tools and materials for checking and evaluating raw milk, as well as finished products, their quality and their conformity with the standard specifications of each product.
- **Management and Marketing Department:** includes the administrative offices, public facilities, restaurants and bathrooms.
- **Water treatment Department:** In order to become usable in the plant, by following a series of mechanical and chemical processes in the following order:
 1. Sedimentation and filtration to dispose of organic matter related to water.
 2. Remove the water shortage resulting from the presence of calcium carbonate, which causes many problems, the most important of which is the deposition of calcareous layer on the surface of the heating panels, which reduces thermal conductivity of these panels, and the water shortage causes the increase in the amount of washing.
 3. Sterilization of chlorinated water to eliminate pathogenic microorganisms.

The supply of hot water and steam necessary for manufacturing and washing operations, through a complete heating circuit, which includes the boiler and its necessary accessories.

- Provision of cold water, necessary for dairy products cooling after being manufactured, or to cool the milk in a full cooling cycle after sterilization.

Technical Requirements Analysis

1.1.1 Land and construction

The project will be constructed on an area of 10,000 m². The following table shows the required land:

Table12 the required land for the project

Land	Area sq.m
The required land	10,000

The required construction works for the project are represented as follows:

Table13 Required Construction Work

Construction Work	Area (sq.m)
Hengar	2,000
Paving	150
Design	2,000

Machinery and Equipment

The following table provides the machinery and equipment needed for the project:

Table 14 Machinery and Equipment

Machinery and equipment	Quantity
Cooling tanks	3
Dairy production line	1
Milk production line	1
Pasteurization machine	3
Filtering and purifying machine	1
Boiler	2
Laboratory devices	-
Processing equipment	1
Generator	1
Heatsink equipment	-
Oven	1
Packaging machine	1
Tubes	-
Stainless steel cabinets and tables	4
Sorting and blending machine	1

Vehicles and Transportation

The following table shows a list of vehicles and means of transportation required for the project:

Table 15 Required Vehicles and Transportation

Vehicles and Transportation	Quantity
Refrigerated truck 3-ton capacity	5
Pick-ups	3

4. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 12.26% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 5 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 16 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 1 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 17 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 18 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	50%
Utilities Expenses from Total Revenues	10%
Maintenance Expenses from Total Revenues	1%
Depreciation Expenses from Total Revenues	1%
Insurance Expenses from Total Assets Value	0.5%
Marketing Expenses from Total Revenues	5%

Table 19 Income Tax Assumptions

Income Tax Assumptions	Value
Average Income Tax in Jordan(1)	20%
Income Tax Deduction(2)	80%
Income Tax after Deduction	5%
Compulsary Reserve Percentage(3)	10%
Other Assumptions	Value
Annual Monthly Salaries have been calculated after multiplying monthly salaries by:(1)	16

Risk Premium	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	9.48%
(3) Market Risk Premium	10.12%
Income Tax Rate	5%
(4) Beta	0.89
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 20 Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premium	%10.12
Income Tax Rate	%5
Bets	1.00
Equity	1,004,802
Loans Value	1,507,203
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%9.01
Loans Value	1,507,203

Loans Percentage	%60
Equity	
Cost of Equity	%16.4
Equity Value	1,004,802
Equity Percentage	%40
Gross	
Project Value	2,512,005
Weighted Average Cost of Capital	%12.05

Capital Expenditures

The estimated cost of the project is 2,512,005 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 21 Land Capital Expenditure

Land	(Jordan Dinar/Meters Square) Cost	Area (m2)	Cost
Required Land	5	10,000	50,000
Gross Cost			50,000

Table 22 Construction Work Capital Expenditure

Construction Works	Cost (JOD per Squared	Area	Gross
Hunger	100	2,000	200,000
paving	12	150	1,800
Design and blueprints	7	2,000	14,000
Other Works			21,580
Total			237,380

Table 23 machinery and equipments Work Capital Expenditure

Machinery and equipments	No.	Cost (JOD per unit)	Gross cost
Cooling tanks	3	10,500	31,500
Dairy production line	1	63,000	63,000
Milk production line	1	81,700	81,700
Pasteurization machine	3	8,400	25,200
Filtering and purifying machine	1	35,000	35,000
Boiler	2	24,500	49,000
Laboratory devices	2	20,000	40,000
Processing equipment	1	28,500	28,500
Generator	1	5,000	5,000
Heatsink equipment	1	3,000	3,000
Oven	1	2,000	2,000
Packaging machine	1	400	400
Tubes	-	3,000	3,000
Stainless steel cabinets and tables	-	-	2,900
Sorting and blending machine	1	4,000	4,000
Other			4,000
Total			378,200

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Table 24 Transporation and Vehicles Capex

Description	No.	Cost (JOD per	Gross cost
Refrigerated truck 3-ton capacity	5	27,200	136,000
Pick-ups	3	23,400	70,200
Total	8		206,200

Table 25 Pre Operating Expenses

Pre-operating expenses	Estimated cost (JD)
Governmental fees	2,500
Transportation	2,500
Marketing	5,000
Training	1,850
Total	11,850

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 26 Working Capital

Working Capital	No. of	%	Estimated Cost
Operating expenses	6	%50	1,182,681
General and administrative expenses	6	%50	18,972
Marketing expenses	6	%50	81,158
Indirect Salaries	6	%50	99,200
Total			1,382,011

Table 27 Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Land	50,000	10%	55,000	%2.2
Construction works	237,380	10%	261,118	%10.4
Machinery & Equipment	378,200	10%	416,020	%16.6
Furniture	18,000	10%	19,800	%0.8
Vehicles	206,200	10%	226,820	%9.0
Pre-Operating Expenses	11,850	10%	13,035	%0.5
Working capital	1,382,011	10%	1,520,212	%60.5
Total (JD)	2,283,641		2,512,005	100%

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (2,512,005) JD, where 60% of the project will be financed through loans (1,507,203) JD, and the remaining (40%) through (Equity) with (1,004,802) JD. The following table summarizes the general structure of the required financing:

Table 28 Sources of Funding

Funding Sources	Amount	Percentage
Equity (Self Financing)	1,004,802	%40
Loans	1,507,203	%60
Total	2,512,005	%100

Use of Fund	Value	Ratio
Capital Expenditures	978,758	%38.96
Pre-Operating Expenses	13,035	%0.52
Working Capital	1,520,212	%60.52
Total	2,512,005	%100.00

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 29 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials and Packaging from Revenues	1,623,168	1,758,432	1,893,696	2,028,960	2,028,960	2,164,224	2,164,224	2,299,488	2,434,752	2,570,016
Utilities from Revenues	324,634	351,686	378,739	405,792	405,792	432,845	432,845	459,898	486,950	514,003
Maintenance from Revenues	32,463	35,169	37,874	40,579	40,579	43,284	43,284	45,990	48,695	51,400
Disposables from Revenues	32,463	35,169	37,874	40,579	40,579	43,284	43,284	45,990	48,695	51,400
Direct Salaries	137,600	143,242	149,115	155,228	161,593	168,218	175,115	182,294	189,769	197,549
Others	215,033	232,370	249,730	267,114	267,750	285,186	285,875	303,366	320,886	338,437
Total	2,365,361	2,556,067	2,747,027	2,938,252	2,945,253	3,137,041	3,144,628	3,337,026	3,529,747	3,722,806

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027)

Table 30 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	2,400	2,472	2,546	2,623	2,701	2,782	2,866	2,952	3,040	3,131
Hospitality	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Stationary and Printing	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Transportation	3,600	3,708	3,819	3,934	4,052	4,173	4,299	4,428	4,560	4,697
Rent	20,000	20,600	21,218	21,855	22,510	23,185	23,881	24,597	25,335	26,095
Training	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Insurance	4,894	5,041	5,192	5,348	5,508	5,673	5,843	6,019	6,199	6,385
Miscellaneous	3,449	3,553	3,659	3,769	3,882	3,999	4,119	4,242	4,370	4,501
Total	37,943	39,081	40,254	41,462	42,705	43,987	45,306	46,665	48,065	49,507

Marketing Expenses

Table 31 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	162,317	175,843	189,370	202,896	202,896	216,422	216,422	229,949	243,475	257,002

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summarizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 32 Expected Annual Salaries and Wages (2027-2018)

Job Title	No.	Monthly salary (JD)	Total monthly	Total annual wages(دينار)
General Manager	1	3,000	3,000	48,000
Financial Manager	1	2,000	2,000	32,000
Quality manager	1	1,500	1,500	24,000
Human resources manager	1	1,250	1,250	20,000
Quality auditors	2	500	1,000	16,000
Accountant	3	500	1,500	24,000
Secretaries	1	450	450	7,200
Human resources staff	2	500	1,000	16,000
security	2	350	700	11,200
Indirect employes	14		12,400	198,400
Factory manager	1	1,500	1,500	24,000
Production line technicians	10	250	2,500	40,000
Logistics Manager	1	1,250	1,250	20,000
Maintenance technicians	2	300	600	9,600
Driver	5	350	1,750	28,000
Logistics staff	2	500	1,000	16,000
Direct employes	21		8,600	137,600

Job Title	No.	Monthly salary (JD)	Total monthly	Total annual wages (دينار)
Total	35		21,000	336,000

Depreciations

The following tables illustrate cost of Capex and Depreciation rates:

Table 33 Capex and Depreciation Expenses

Equipment costs and capital expenditure, annual consumption rates	Cost (JOD)	Consumption ratio	Annual additions
Land	55,000	%0.0	%0.0
Construction Works	261,118	%5.0	%0.0
Machinaries	416,020	%10.0	%2.0
Furniture	19,800	%10.0	%5.0
Vehicles	226,820	%10.0	%0.0
Gross Total (JOD)	978,758		

Table 34 Depreciations and Additions on Construction Works

Summary of capital costs and additions and consumption annually	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	978,758	988,068	997,595	1,007,343	1,017,318	1,027,528	1,037,978	1,048,675	1,059,625	1,070,837
Total Depreciation Expenses	79,320	80,251	81,204	82,178	83,176	84,197	85,242	86,312	87,407	88,528
Total Accumulated Depreciation	79,320	159,571	240,774	322,953	406,129	490,326	575,568	661,879	749,286	837,814
Total Additions	0	9,310	9,526	9,748	9,976	10,210	10,450	10,697	10,951	11,211
Total Net Book Values	899,438	828,498	756,820	684,390	611,190	537,202	462,410	386,796	310,340	233,023

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 35 Loan Details

Loan value	1,507,203
Annual Interest Rate	9.48%
Loans Period	5
Loan Starts at year:	2017
Annual Payment	392,328
Number of Payments	5

The following table shows the loan payments:

year	Annual Payment	Interest	Capital	The remainder of the loan
2017				1,507,203
2018	392,328	142,883	249,446	1,257,757
2019	392,328	119,235	273,093	984,664
2020	392,328	93,346	298,982	685,682
2021	392,328	65,003	327,326	358,356
2022	392,328	33,972	358,356	0

Income Statement

Through all the information and financial analysis of the project's financial expenses have statment income for the years (2018 – 2027) as shown in the following table:

Table 36 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	3,246,336	3,516,864	3,787,392	4,057,920	4,057,920	4,328,448	4,328,448	4,598,976	4,869,504	5,140,032
Gross Operating Revenues	3,246,336	3,516,864	3,787,392	4,057,920	4,057,920	4,328,448	4,328,448	4,598,976	4,869,504	5,140,032
Operating Expenses	(2,365,361)	(2,556,067)	(2,747,027)	(2,938,252)	(2,945,253)	(3,137,041)	(3,144,628)	(3,337,026)	(3,529,747)	(3,722,806)
Gross Operating Profit	880,975	960,797	1,040,365	1,119,668	1,112,667	1,191,407	1,183,820	1,261,950	1,339,757	1,417,226
<i>Gross Profit Percentage</i>	<i>%27</i>	<i>%27</i>	<i>%27</i>	<i>%28</i>	<i>%27</i>	<i>%28</i>	<i>%27</i>	<i>%27</i>	<i>%28</i>	<i>%28</i>
Salaries and Benefits (Indirect Staff)	(198,400)	(206,534)	(215,002)	(223,817)	(232,994)	(242,547)	(252,491)	(262,843)	(273,620)	(284,838)
General and Administraive Expenses	(37,943)	(39,081)	(40,254)	(41,462)	(42,705)	(43,987)	(45,306)	(46,665)	(48,065)	(49,507)
Markeing Expenses	(162,317)	(175,843)	(189,370)	(202,896)	(202,896)	(216,422)	(216,422)	(229,949)	(243,475)	(257,002)
Pre Operating Expenses	(13,035)									
Gross Indirect Expenses	(411,695)	(421,459)	(444,626)	(468,175)	(478,595)	(502,956)	(514,220)	(539,457)	(565,160)	(591,347)
Income before Interest, Depreciation and Tax	469,280	539,338	595,739	651,493	634,071	688,451	669,601	722,493	774,597	825,879
Fixed Assets Depreciations	(79,320)	(80,251)	(81,204)	(82,178)	(83,176)	(84,197)	(85,242)	(86,312)	(87,407)	(88,528)
Income before Tax and Interests	389,960	459,087	514,535	569,314	550,896	604,254	584,359	636,182	687,190	737,351
Bank Interests	(142,883)	(119,235)	(93,346)	(65,003)	(33,972)	(0)	(0)	(0)	(0)	(0)
Income Before Tax	247,077	339,852	421,189	504,312	516,923	604,254	584,359	636,182	687,190	737,351
Income Tax	(12,354)	(16,993)	(21,059)	(25,216)	(25,846)	(30,213)	(29,218)	(31,809)	(34,359)	(36,868)
Net Profit	234,723	322,859	400,130	479,096	491,077	574,042	555,141	604,372	652,830	700,484

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
<i>Net Profit Percentage</i>	<i>%7</i>	<i>%9</i>	<i>%11</i>	<i>%12</i>	<i>%12</i>	<i>%13</i>	<i>%13</i>	<i>%13</i>	<i>%13</i>	<i>%14</i>
Compulsory Reserves	(23,472)	(32,286)	(40,013)	(47,910)	(49,108)	(57,404)	(55,514)	(60,437)	(65,283)	(70,048)
Retained Earnings	211,251	501,824	861,941	1,293,127	1,735,097	2,251,734	2,751,361	3,295,296	3,882,843	4,513,279

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below

Table 37 Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	234,723	322,859	400,130	479,096	491,077	574,042	555,141	604,372	652,830	700,484
Bank Interests	142,883	119,235	93,346	65,003	33,972	0	0	0	0	0
Depreciation	79,320	80,251	81,204	82,178	83,176	84,197	85,242	86,312	87,407	88,528
Total Operating Cashflows before Additions to Working Capital	456,926	522,345	574,679	626,277	608,225	658,239	640,383	690,684	740,237	789,012
Inventory (Increase/Decrease)	(135,264)	(11,272)	(11,272)	(11,272)	0	(11,272)	0	(11,272)	(11,272)	(11,272)
Accounts Receivables (Increase/Decrease)	(270,528)	(22,544)	(22,544)	(22,544)	0	(22,544)	0	(22,544)	(22,544)	(22,544)
Accounts Payables (Increase/Decrease)	185,548	13,860	13,924	13,990	1,750	14,131	1,897	14,283	14,364	14,449
Working Capital (Increase/Decrease)	(220,244)	(19,956)	(19,892)	(19,826)	1,750	(19,685)	1,897	(19,533)	(19,452)	(19,367)

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Cashflows from Operating Activities	236,682	502,390	554,787	606,451	609,975	638,553	642,279	671,151	720,785	769,644
Cashflows from Investments Activities										
Fixed Assets (Procurement)	(978,758)	(9,310)	(9,526)	(9,748)	(9,976)	(10,210)	(10,450)	(10,697)	(10,951)	(11,211)
Net Cashflows from Investments Activities	(978,758)	(9,310)	(9,526)	(9,748)	(9,976)	(10,210)	(10,450)	(10,697)	(10,951)	(11,211)
Cashflows from Financing Activities										
Capital	1,004,802									
Loan Amortization	(249,446)	(273,093)	(298,982)	(327,326)	(358,356)	0	0	0	0	0
Bank Interest Rate	(142,883)	(119,235)	(93,346)	(65,003)	(33,972)	(0)	(0)	(0)	(0)	(0)
Loans	1,507,203	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	2,119,676	(392,328)	(392,328)	(392,328)	(392,328)	0	0	0	0	0
Net (Increase/Decrease) in Cash	1,377,601	100,751	152,933	204,375	207,671	628,344	631,829	660,455	709,835	758,433
Cashflows at the Beginning of Period	0	1,377,601	1,478,352	1,631,284	1,835,659	2,043,331	2,671,675	3,303,504	3,963,959	4,673,793
Cashflows at the End of Period	1,377,601	1,478,352	1,631,284	1,835,659	2,043,331	2,671,675	3,303,504	3,963,959	4,673,793	5,432,226

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 38 Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current Assets										
Cash	1,377,601	1,478,352	1,631,284	1,835,659	2,043,331	2,671,675	3,303,504	3,963,959	4,673,793	5,432,226
Inventory	135,264	146,536	157,808	169,080	169,080	180,352	180,352	191,624	202,896	214,168
Accounts Receivable	270,528	293,072	315,616	338,160	338,160	360,704	360,704	383,248	405,792	428,336
Total current Assets	1,783,393	1,917,960	2,104,708	2,342,899	2,550,571	3,212,731	3,844,560	4,538,831	5,282,481	6,074,730
Non Current Assets										
Fixed Assets (net)	899,438	828,498	756,820	684,390	611,190	537,202	462,410	386,796	310,340	233,023
Total Non Current Assets	899,438	828,498	756,820	684,390	611,190	537,202	462,410	386,796	310,340	233,023
Total Assets	2,682,831	2,746,457	2,861,529	3,027,289	3,161,760	3,749,933	4,306,970	4,925,626	5,592,821	6,307,754
Liabilities	-	-	-	-	-					
Current Liabilities										
Paybles	185,548	199,409	213,333	227,323	229,073	243,204	245,101	259,384	273,749	288,197
Remaining amount of Loan	273,093	298,982	327,326	358,356	(0)	(0)	(0)	(0)	(0)	0
Total current Liabilities	458,641	498,391	540,659	585,679	229,073	243,204	245,101	259,384	273,749	288,197
Non Current Liabilities										
Long Terms Loans	984,664	685,682	358,356	0	0	0	0	0	0	0
Total Long Term Liabilities	984,664	685,682	358,356	0	0	0	0	0	0	0
Total Liabilities	1,443,305	1,184,073	899,015	585,679	229,073	243,204	245,101	259,384	273,749	288,197

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Owners Equity										
Shareholders Contributions	1,004,802	1,004,802	1,004,802	1,004,802	1,004,802	1,004,802	1,004,802	1,004,802	1,004,802	1,004,802
Statutory Reseve	23,472	55,758	95,771	143,681	192,789	250,193	305,707	366,144	431,427	501,475
Retained Profits	211,251	501,824	861,941	1,293,127	1,735,097	2,251,734	2,751,361	3,295,296	3,882,843	4,513,279
Total Equity	1,239,525	1,562,384	1,962,514	2,441,610	2,932,687	3,506,729	4,061,869	4,666,242	5,319,072	6,019,556
Total Liabilities and Equity	2,682,831	2,746,457	2,861,529	3,027,289	3,161,760	3,749,933	4,306,970	4,925,626	5,592,821	6,307,754

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 39 Free Net Cash flows Table

Net Cashflows	Pre- Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final value
Net Free Cashflows	(2,512,005)	249,717	493,079	545,261	596,703	600,000	628,344	631,829	660,455	709,835	758,433	9,796,441
Discount Factor	1.00	0.71	0.63	0.57	0.51	0.45	0.40	0.36	0.32	0.29	0.26	0.26
Net Present Value for Cash Flows	(2,512,005)	177,498	312,784	308,684	301,474	270,536	252,844	226,901	211,671	203,029	193,598	2,500,642

Table below (40) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 40 Payback Period

Payback Period	Pre- Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	(2,512,005)	249,717	493,079	545,261	596,703	600,000	628,344	631,829	660,455	709,835	758,433
Project Value Returned	2,512,005	2,262,287	1,769,208	1,223,947	627,243	27,244	(601,100)	(1,232,930)	(1,893,384)	(2,603,219)	(3,361,652)
Payback Period (Year)	5	1	1	1	1	1	0	0	0	0	0

Table 41 Financial Analysis Results

indicator	Value
Weighted Average Cost of Capital (WACC)	%12.05
Net Present Value for Cashflows	2,447,658
Payback Period	5
Internal Rate of Return	%26.13

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table 42 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%26.13	5	12.05%
Revenues declined by 10%	%23.45	5	12.05%
Operating Expenses Increased by 10%	%14.63	8	12.05%

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 2,447,658 Jordan Dinars considering that the project provides 35 Job Opportunities for the governorate residents.